



CHEMISTRY

Paper 3 Free Response

9647/03

16 September 2014

2 hours

Candidates answer on separate paper.

Additional Materials: Answer Paper
 Data Booklet
 Cover Page

READ THESE INSTRUCTIONS FIRST

Write your name, class and exam index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer any **four** questions.

A Data Booklet is provided.

The use of an approved scientific calculator is expected, where appropriate.

You are reminded of the need for good English and clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

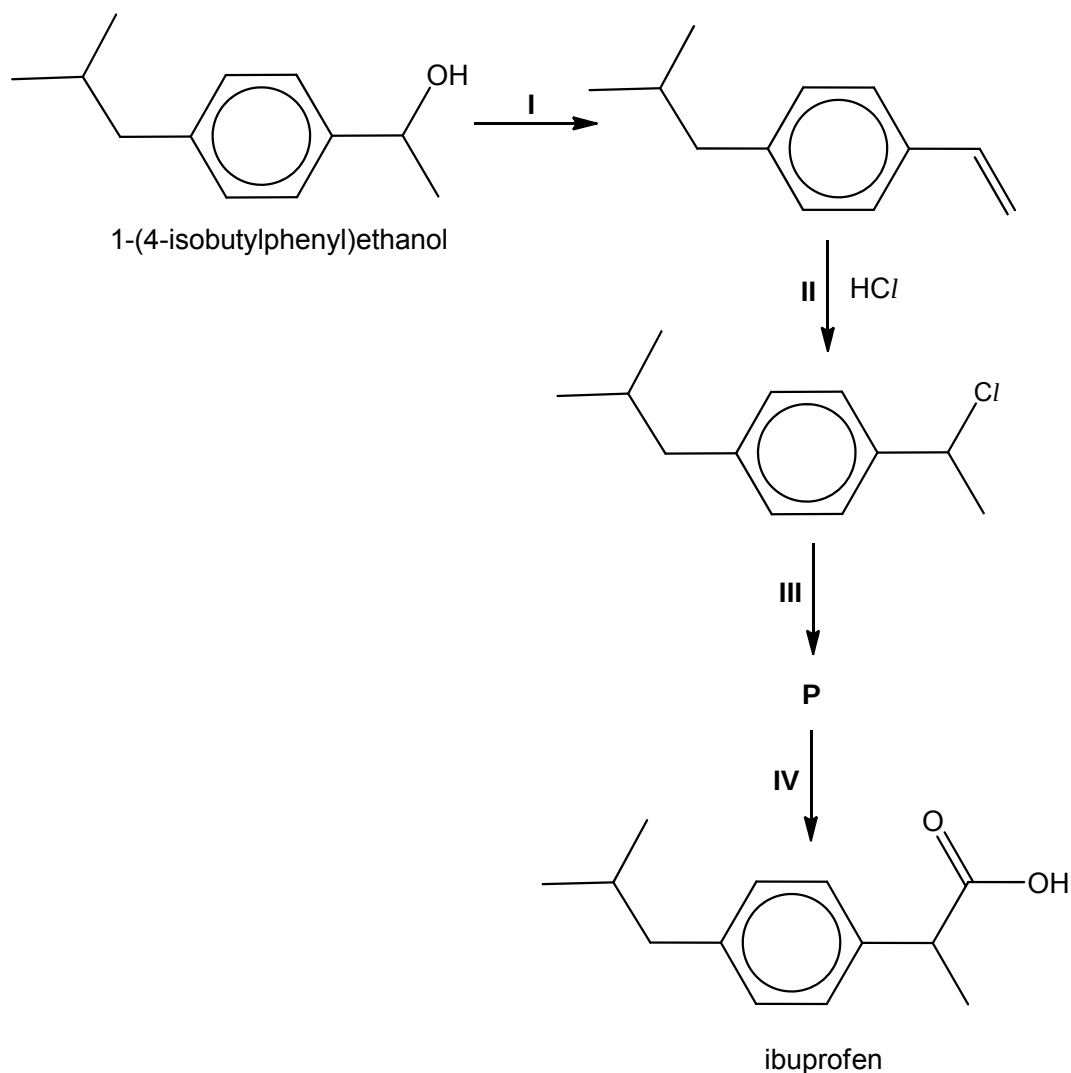
At the end of the examination, fasten all your work securely together.

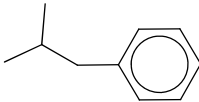
This document consists of **12** printed pages and **2** blank pages.

Answer any **four** questions.

- 1 (a) Ibuprofen is a non-steroidal anti-inflammatory drug (NSAID) and has many important applications in the pharmaceutical industry. It is used commonly to reduce inflammation, pain and fever.

Ibuprofen can be synthesised from 1-(4-isobutylphenyl)ethanol by the following route.

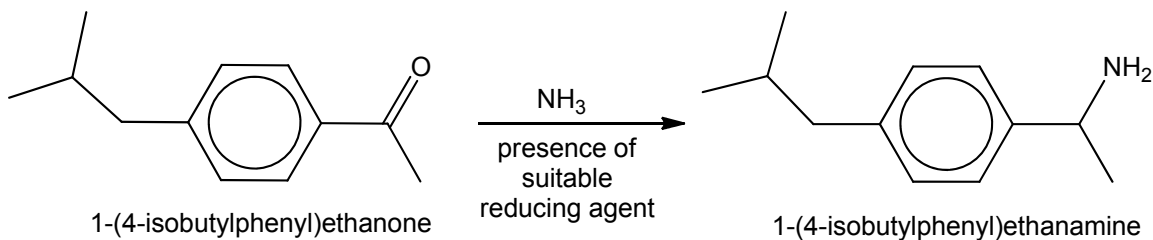


- (i) Draw the structure of compound **P** and state the reagents and conditions for reactions **I**, **III** and **IV**.
- (ii) Describe the mechanism for reaction **II**. In your answer, use curly arrows to show the movement of electrons and indicate clearly all relevant charges.
- [You may denote  as R.]
- (iii) By considering the mechanism you have drawn in **(a)(ii)**, explain why the resulting product formed in reaction **II** is optically inactive.
- (iv) Suggest why ibuprofen is not very soluble in water.

[8]

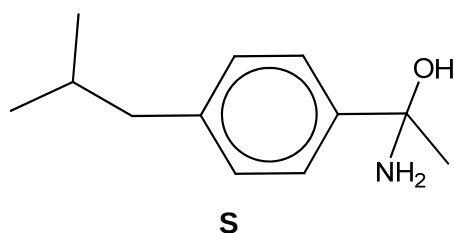
- (b) Like ibuprofen, 1-(4-isobutylphenyl)ethanamine is widely used in the pharmaceutical industry as an intermediate in proteomics research on the study of the interactions between drugs and proteins in the body.

It can be formed through a reductive amination process as shown below.



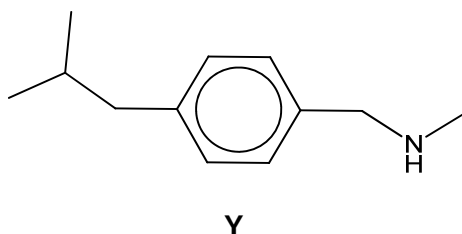
The reaction occurs in three steps.

- The first step begins with a nucleophilic addition, where the ammonia nucleophile attacks 1-(4-isobutylphenyl)ethanone, which results in intermediate **S**.

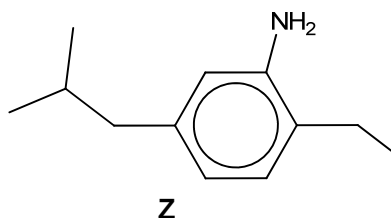


- S** subsequently undergoes dehydration followed by reduction to give 1-(4-isobutylphenyl)ethanamine.

In a similar reaction, a carbonyl compound **W** undergoes reductive amination with an amine **X** (instead of ammonia) to give compound **Y**, a structural isomer of 1-(4-isobutylphenyl)ethanamine.



- Suggest the structural formulae of compounds **W** and **X**.
- Compound **Z** is another structural isomer of 1-(4-isobutylphenyl)ethanamine.

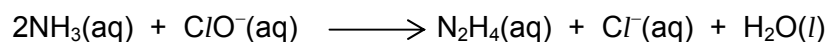


Suggest how the basicity of 1-(4-isobutylphenyl)ethanamine might compare to that of compound **Z**. Give a reason for your answer

- Suggest a chemical test that could be used to distinguish between 1-(4-isobutylphenyl)ethanamine and **Z**.

[6]

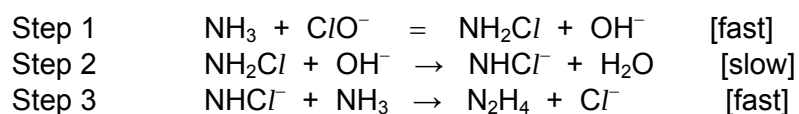
- 1 (c) In aqueous medium, ammonia reacts with hypochlorite, ClO^- , to produce hydrazine. Hydrazine, N_2H_4 , is commonly used as a rocket propellant in spacecrafts.



A chemist studied the kinetics of the above reaction at different initial concentrations of NH_3 and ClO^- . He obtained the following table of results.

experiment number	initial $[\text{NH}_3(\text{aq})]$ / mol dm^{-3}	initial $[\text{ClO}^-(\text{aq})]$ / mol dm^{-3}	initial rate / $\text{mol dm}^{-3} \text{s}^{-1}$
1	1.0×10^{-4}	1.0×10^{-4}	4.0×10^{-12}
2	1.0×10^{-4}	0.5×10^{-4}	2.0×10^{-12}
3	3.0×10^{-4}	4.0×10^{-4}	4.8×10^{-11}

- (i) Determine the order of reaction with respect to $[\text{NH}_3]$ and $[\text{ClO}^-]$. Explain your reasoning.
- (ii) Hence write the rate equation for the reaction, and calculate a value for the rate constant. Include units in your answer.
- (iii) The above reaction is thought to proceed via the following reaction mechanism.

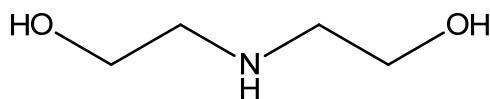


Show how this reaction mechanism is consistent with your experimentally determined rate equation in (c)(ii).

[6]

[Total : 20]

2. Diethanolamine (DEA), $(\text{HOCH}_2\text{CH}_2)_2\text{NH}$, is a monoprotic base which is used as a neutralising agent in agriculture. It has a K_b value of $3.98 \times 10^{-5} \text{ mol dm}^{-3}$.



DEA

The pH of a 20.0 cm^3 solution of DEA is 11.2. This solution is exactly neutralised by 25.00 cm^3 of $0.050 \text{ mol dm}^{-3}$ hydrochloric acid.

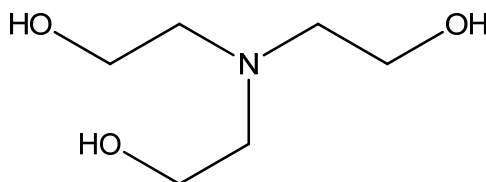
- (a) (i) DEA is a *weak Bronsted base*. What do you understand by the terms in italics?
- (ii) Show, by means of calculations, that DEA is indeed a weak base.
- (iii) DEA behaves like ammonia. Write the K_b expression for DEA.
(You should use R_2NH as the chemical representation of DEA in all your answers.)
- (iv) In the above titration, the pH at equivalence point is not 7. Write an equation to account for this observation.

[5]

- (b) To another 20.0 cm^3 solution of DEA, 10.00 cm^3 of $0.050 \text{ mol dm}^{-3}$ hydrochloric acid was added.
- (i) Give the formulae of the species present in the resultant solution.
- (ii) Calculate the pH of the resultant solution.
- (iii) Explain the effect on the pH of the resultant solution when one drop of NaOH(aq) is added.

[4]

- (c) Another compound in the ethanolamine family is triethanolamine (TEA), which is also used in the agricultural chemical field.



TEA

- (i) Predict which compound, DEA or TEA, is expected to have a higher boiling point than the other. Explain your answer in terms of structure and bonding.
- (ii) Both DEA and TEA are highly soluble in water. Draw a diagram to illustrate how a molecule of DEA interacts with a molecule of water.

[3]

- (d) Other than the pH conditions of soil, presence of excessive amounts of metallic ions such as cadmium is also becoming a concern in agricultural field. Cadmium ions will not only affect agricultural yield but may also lead to toxic poisoning in humans.

A soil sample was analysed for its suitability in vegetable cultivation.

- (i) The pH of the soil sample at 25 °C is 9.6. Calculate the concentration of cadmium ions in g dm^{-3} , given that the K_{sp} of $\text{Cd}(\text{OH})_2$ is $2.5 \times 10^{-14} \text{ mol}^3 \text{ dm}^{-9}$.
- (ii) Deduce whether the soil sample will produce vegetable that is safe for human consumption, given that the safe concentration of cadmium ions in agricultural soil is between 0.01 to $0.7 \mu\text{g cm}^{-3}$. [$1 \mu\text{g} = 1 \times 10^{-6} \text{ g}$]

[3]

- (e) When one mole of DEA is treated with one mole of ClCH_2COCl , a neutral compound, **F** is formed. Warming one mole of compound **F** with one mole of sodium metal produces a cyclic compound **G**, $\text{C}_6\text{H}_{11}\text{NO}_3$. Heating **G** with aluminium oxide gives compound **H** which decolourises bromine water.

Suggest the structures of **F**, **G** and **H**. State the type of reactions which occur.

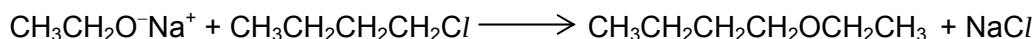
[5]

[Total : 20]

- 3 The compounds of Group VII elements are valuable commodities due to their wide ranging applications and uses. The organic halides are important precursors for the synthesis of many organic compounds while the oxyanions of the Group VII elements are well known for their powerful oxidising properties.

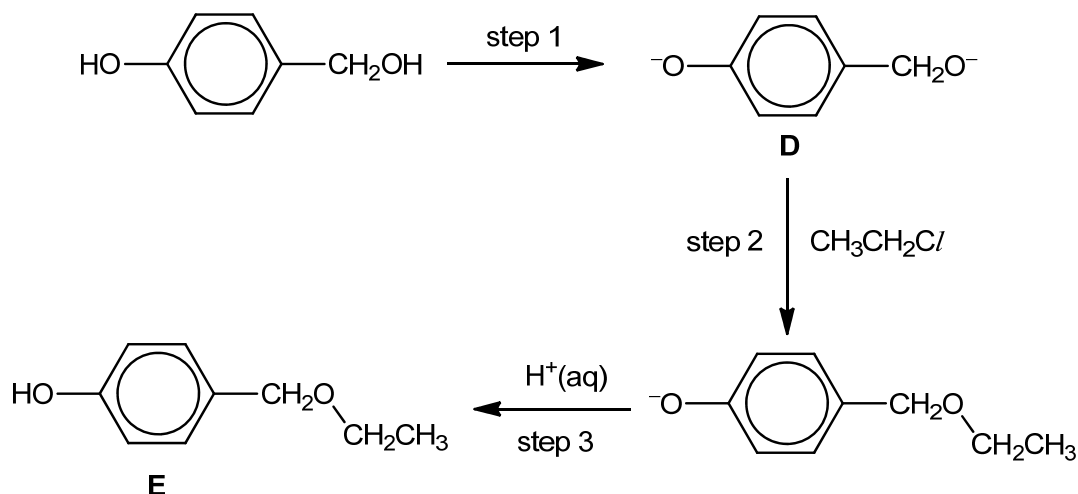
- (a) Williamson ether synthesis involves the preparation of ethers, $R-O-R'$ (where R and R' are alkyl or aryl), from the reaction between alkoxides and alkyl halides. This method is suitable for the preparation of a wide variety of unsymmetric ethers.

One example is the reaction between sodium ethoxide ($\text{CH}_3\text{CH}_2\text{O}^-\text{Na}^+$) and 1-chlorobutane. The kinetics of the reaction is found to be first order with respect to both ethoxide ion and 1-chlorobutane.



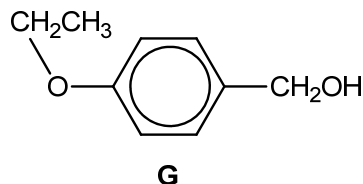
Describe the mechanism of the above reaction, using $\text{CH}_3\text{CH}_2\text{O}^-$ as the nucleophile. [2]

- (b) The following reaction scheme shows the synthesis of compound **E** via Williamson ether synthesis.



- (i) Suggest a reagent for step 1.
- (ii) In the above reaction, compound **E** is the only organic product formed. Compound **G**, as shown below, is **not** formed as the phenoxide group in the intermediate **D** is a weak nucleophile.

Explain why phenoxide is a weak nucleophile.



- (iii) The same reaction in (b) was repeated using iodoethane instead of chloroethane in step 2. With the use of the *Data Booklet*, suggest and explain the effect of this change on the rate of reaction in step 2. [4]

(c) Use of the Data Booklet is relevant to this part question.

Potassium chlorate(V), KClO_3 , is formed by mixing the products from the electrolysis of concentrated aqueous KCl solution.

- (i) Predict, with reasons, the products formed at each electrode in the electrolysis of concentrated KCl solution.
- (ii) Describe, with the aid of a balanced equation, how KClO_3 can be formed from the products predicted in (c)(i).

The standard enthalpy changes, ΔH° , of different chlorine-containing species are tabulated below.

half-equation	$\Delta H^\circ / \text{kJ mol}^{-1}$
$2\text{ClO}_3^- + 12\text{H}^+ + 10\text{e}^- \rightarrow \text{Cl}_2 + 6\text{H}_2\text{O}$	-986
$\text{Cl}_2 + 2\text{e}^- \rightarrow 2\text{Cl}^-$	-334

- (iii) Write a half-equation for the reduction of ClO_3^- to Cl^- in an acidic medium. Construct an energy cycle with the above data provided, and hence calculate ΔH° for this reduction.
- (iv) ΔG° and E° are related by the following equation.

$$\Delta G^\circ = -zFE^\circ$$

where ΔG° is the standard Gibbs free energy change in **joules per mole**, E° is the standard electrode potential, z is the number of moles of electrons transferred during the redox reaction and F is the Faraday constant.

Given that ΔG° for the reduction of ClO_3^- to Cl^- is -841 kJ mol^{-1} , use your equation in (c)(iii) to decide on a suitable value for z , and hence calculate the E° for this reduction.

[10]

- (d) An acidified solution of the salt KClO_x will oxidise $\text{Fe}^{2+}(\text{aq})$ to $\text{Fe}^{3+}(\text{aq})$ quantitatively, the chlorine being reduced to $\text{Cl}^-(\text{aq})$.

When 0.302 g of the salt KClO_x was reacted with $0.540 \text{ mol dm}^{-3} \text{ Fe}^{2+}(\text{aq})$ in the presence of $\text{H}^+(\text{aq})$, 21.0 cm^3 of $\text{Fe}^{2+}(\text{aq})$ was needed for complete reaction.

- (i) Calculate, in terms of x , the reacting mole ratio of ClO_x^- and Fe^{2+} .
- (ii) Hence calculate the value of x .

[4]

[Total : 20]

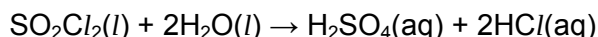
- 4 Sulfuryl chloride, SO_2Cl_2 , is commonly confused with thionyl chloride, SOCl_2 . The properties of these two sulfur oxychlorides are quite different: SO_2Cl_2 is a source of chlorine while SOCl_2 is a source of chloride ions for various organic reactions.

(a) When heated, sulfuryl chloride decomposes endothermically as follows:



In an experiment, 1.0 mol of SO_2Cl_2 vapour was heated in a closed 3.60 dm^3 flask at 400 K until equilibrium had been established. The flask was then rapidly cooled to liquefy SO_2Cl_2 .

After removing gaseous SO_2 and Cl_2 , excess water was then carefully added to the liquid SO_2Cl_2 , causing the following reaction to occur.

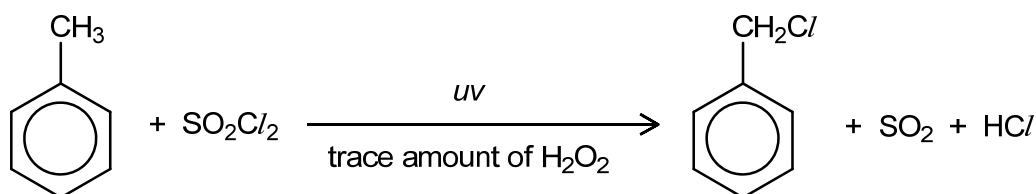


The resulting solution was made up to 250 cm^3 in a standard volumetric flask. 25.0 cm^3 of this solution was titrated with 5.00 mol dm^{-3} NaOH and required 38.50 cm^3 for complete neutralisation.

- Write an expression for K_c for the equilibrium above, stating the units.
- Calculate the equilibrium amount of SO_2Cl_2 , SO_2 , and Cl_2 in the original equilibrium at 400 K. Hence calculate a value for the K_c for the equilibrium.
- Suggest, with a reason, how the position of equilibrium might alter when the temperature of the original equilibrium mixture is increased to 500 K.

[7]

- (b) SO_2Cl_2 is widely used as a reagent in chlorination of the hydrocarbons. Such reactions occur under free radical conditions using H_2O_2 as an initiator. For example, methylbenzene undergoes free radical substitution with SO_2Cl_2 to give chlorophenylmethane.



The mechanism of this reaction is as follows.

- The initiation step is unique as it involves two steps.

In the first step, there is an initial homolytic breaking of the peroxide bond $\text{HO}-\text{OH}$ in hydrogen peroxide forming two hydroxyl radicals.

The hydroxyl radicals then combine with SO_2Cl_2 to give SO_2 , HOCl and a $\text{Cl}^\bullet$ radical.

- The $\text{Cl}^\bullet$ radical then reacts with methylbenzene in the propagation step to form the chlorophenylmethane and regenerating the hydroxyl radicals.
- The chain reaction terminates when two radicals combine to give stable compounds.

Use the information above to describe the mechanism in the initiation, propagation and the termination steps for the reaction between methylbenzene and SO_2Cl_2 in presence of H_2O_2 . You are advised to show clearly how each step proceeds and indicate any unpaired electron by a dot ($^\bullet$).

[4]

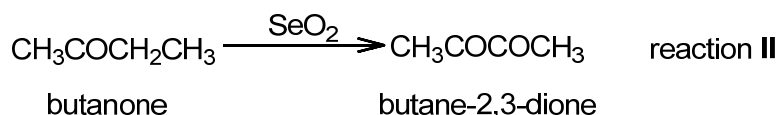
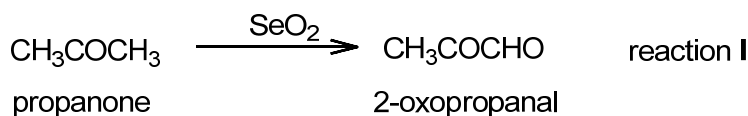
- (c) Selenium, Se, an element below sulfur, is an important precursor in many of the organic synthesis as well. It occurs in nature as a mixture of six isotopes having the relative abundances given below:

Nucleon number	% Abundance
74	0.9
76	9.0
77	7.6
78	23.5
80	49.8
82	9.2

Calculate the relative atomic mass, A_r , of selenium to three significant figures.

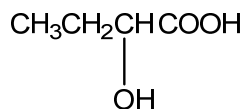
[1]

- (d) In 1932, Riley and friends published their findings on the use of SeO_2 to introduce carbonyl functional group into ketones. Below are some examples of the Riley reaction.



Selenium was produced at the end of each reaction.

- (i) Suggest the role of selenium dioxide in these reactions.
- (ii) Devise a three-step synthesis of 2-hydroxybutanoic acid from butanone, $\text{CH}_3\text{CH}_2\text{COCH}_3$, using the Riley reaction as one of the steps.



2-hydroxybutanoic acid

Suggest reagents and conditions for each step, and draw the structural formulae of the intermediates formed.

[5]

- (e) The sublimation of solid selenium dioxide is spontaneous only at temperatures above 610 K. Predict, with reasons, the sign of ΔH and ΔS for the sublimation of solid SeO_2 and explain why the sublimation process is only spontaneous at such high temperatures.

[3]

[Total : 20]

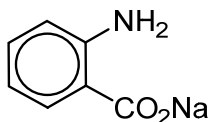
- 5 (a) Iron can exist in oxidation states +2, +3 and +6, whereas the only oxidation state of calcium is +2. Explain why this is so. [2]
- (b) Iron(III) chloride is a dark brown solid which dissolves in water to give an acidic solution. This solution is often used in the electronics industry to etch the copper used in printed circuit boards.
- (i) Explain, with the aid of a chemical equation, why aqueous iron(III) chloride is acidic.
- (ii) Use the *Data Booklet* to explain why iron(III) chloride etches, i.e. dissolves, copper. Write a balanced equation for this reaction. [4]
- (c) When solutions of $\text{Fe}^{3+}(\text{aq})$ and $\text{I}^{-}(\text{aq})$ are mixed, an equilibrium is set up.
- (i) Write a balanced equation for this equilibrium.
- (ii) Use the *Data Booklet* to suggest how the position of equilibrium in (c)(i), involving Fe^{3+} and I^{-} , would change in the presence of $\text{CN}^{-}(\text{aq})$ ions. Explain your answer. [3]
- (d) Solid iron(III) chloride sublimes at 306°C . When 1.00 g of the solid was placed in an evacuated vessel of volume 1.00 dm^3 at 500°C , the pressure produced was 26.3 kPa.
- (i) Calculate the apparent relative molecular mass of the iron(III) chloride under these conditions.
- (ii) Assuming that, in the vapour phase, iron(III) chloride partially exists as a dimer, calculate the percentage of the dimer under these conditions.
- (iii) Draw the structure of the dimer, illustrating the types of bonding present. [3]
- (e) Haemoglobin (Hb) is the iron-containing protein which is responsible for transporting oxygen around the body of mammals.
- With reference to the haemoglobin molecule, explain what is meant by the term *quaternary structure of protein*, clearly stating the types of interactions involved. [2]
- (f) One molecule of haemoglobin can bind up to four molecules of oxygen to form oxyhaemoglobin, as described by the following equilibrium.
- $$\text{Hb}(\text{aq}) + 4\text{O}_2(\text{g}) = \text{Hb}(\text{O}_2)_4(\text{aq})$$
- (i) State the type of bonding between oxygen and the iron centre in the oxyhaemoglobin molecule.
- (ii) In terms of the above equilibrium, explain how oxygen is released from oxyhaemoglobin in the bloodstream and supplied to body cells which require oxygen for essential metabolic processes. [2]

- (g) Haemocyanin is a copper-containing oxygen transport molecule in horseshoe crabs. Both oxygenated and deoxygenated forms of haemocyanin contain copper ions. Haemocyanin is colourless when deoxygenated and blue when it is exposed to oxygen.

Suggest the oxidation states of copper in the oxygenated and deoxygenated forms of haemocyanin. Hence explain the difference in colour observed.

[3]

(h)



Sodium 2-aminobenzoate

Sodium 2-aminobenzoate is a bidentate ligand that can react with $\text{Cu}^{2+}(\text{aq})$ to give a six-coordinate complex containing 1:1 molar ratio of the copper ion to the 2-aminobenzoate ligand.

Draw the structure of this complex, showing the geometry around the copper ion.

[1]

[Total : 20]

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